Topic 5 — Continuous Probability Distributions

Continuous Random Variables

Recall: a random variable is a function $X : \Omega \to \mathbb{R}$, and its CDF is $F_X(x) := \mathbb{P}(X \leq x)$, with $0 \leq F_X(x) \leq 1$ and $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$.

So far we discussed the discrete case, $X : \Omega \to \{x_1, x_2, x_3, \dots\}$. Often we need a continuous range of values instead: $X : \Omega \to \mathbb{R}$, or $X : \Omega \to [a, b]$, etc.

Definition

A random variable X is called **continuous** if there exists a function $f_X : \mathbb{R} \to [0, \infty)$, called the **probability density function (PDF)** of X, such that

$$\mathbb{P}(a < X \leq b) = \int_a^b f_X(x)\,\mathrm{d}x.$$

Properties of the PDF

Remark

- $f_X(x) \geq 0$
- $\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1$
- $F_X(x) = \int_{-\infty}^{x} f_X(y)\,\mathrm{d}y$
- $f_X(x) = \frac{d}{dx} F_X(x) = F_X'(x)$

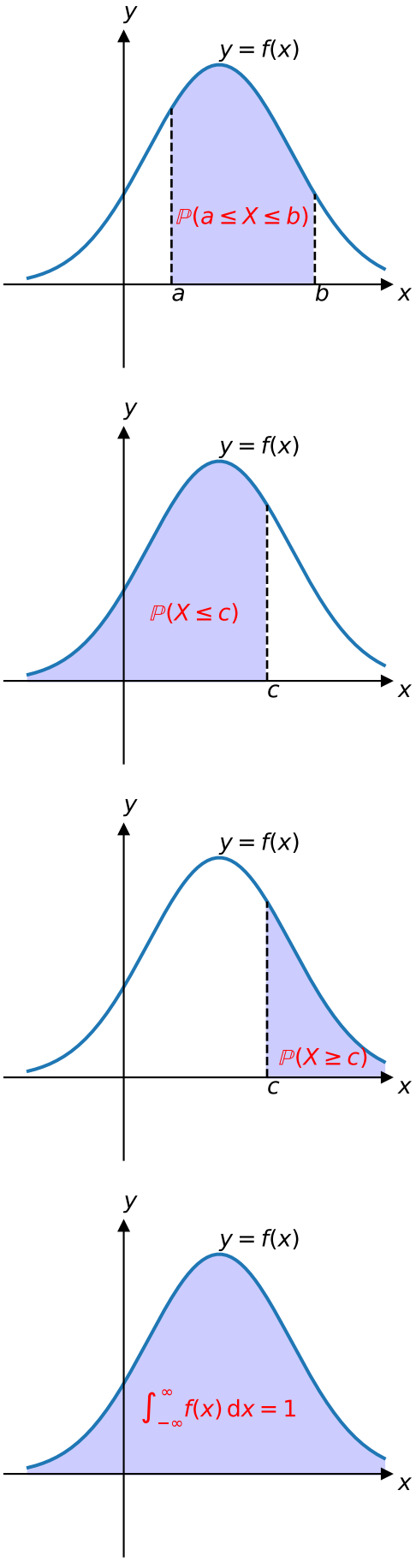


Figure 1: Visual representation of probabilities for a continuous random variable

Zero-Probability Points and Useful Formulas

Remark

For a continuous random variable, $\mathbb{P}(X = a) = 0$ for all $a \in \mathbb{R}$, since $\int_a^a f_X(x)\,\mathrm{d}x = 0$. Hence

$$\mathbb{P}(a \le X \le b) = \mathbb{P}(a \le X < b) = \mathbb{P}(a < X \le b) = \mathbb{P}(a < X < b) = \int_a^b f_X(x)\,\mathrm{d}x.$$

Remark

$$\mathbb{P}(X \le c) = \mathbb{P}(X < c) = F_X(c) = \int_{-\infty}^{c} f_X(x)\,\mathrm{d}x, \qquad \mathbb{P}(X \ge c) = \mathbb{P}(X > c) = 1 - F_X(c) = \int_c^{\infty} f_X(x)\,\mathrm{d}x$$

Expected Value and Variance

Definition

If $X : \Omega \to \mathbb{R}$ is a continuous random variable, its **expected value** is

$$\mathbb{E}(X) := \int_{-\infty}^{\infty} x \cdot f_X(x)\,\mathrm{d}x$$

(provided the integral is finite).

Remark

More generally, for $g : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x)\,\mathrm{d}x, \qquad \text{in particular} \qquad \mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x)\,\mathrm{d}x.$$

Definition

The **variance** of X is $\mathrm{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \geq 0$. For a continuous random variable,

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} x^2 \cdot f_X(x)\,\mathrm{d}x - \left(\int_{-\infty}^{\infty} x \cdot f_X(x)\,\mathrm{d}x\right)^2.$$

The Uniform Distribution

Definition

The **uniform distribution** is a continuous probability distribution where all outcomes within a specified interval $[a, b]$ are equally likely.

Since f_X is constant, say $=$ const, on $[a, b]$ and 0 outside, and the total area must be 1:

$$1 = \int_{-\infty}^{\infty} f_X(x)\,dx = \underbrace{\int_{-\infty}^{a}}_{=0} + \int_a^b \text{const}\,dx + \underbrace{\int_b^{\infty}}_{=0} = \text{const} \cdot (b-a) \implies \text{const} = \frac{1}{b-a}.$$

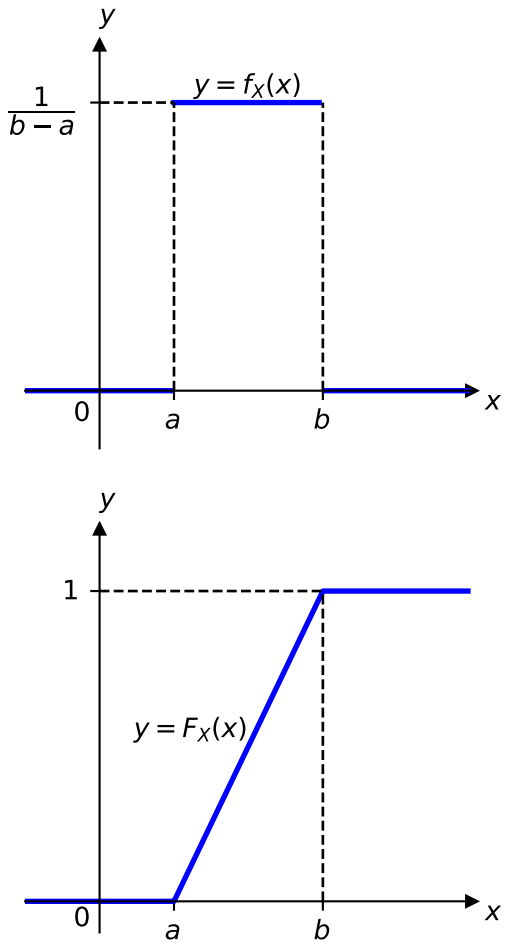


Figure 2: Graphs of PDF and CDF of $X \sim U(a, b)$.

Definition

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise,} \end{cases} \qquad F_X(x) = \begin{cases} 0, & x < a \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b. \end{cases}$$

Notation: $X \sim U(a, b)$.

Indeed, $\mathbb{E}(X) = \displaystyle\int_a^b x \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{2(b-a)}(b^2 - a^2) = \frac{a+b}{2}.$

Definition

$$\mathbb{E}(X) = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

Remark

If $[c,d] \not\subset [a,b]$, one must intersect with $[a,b]$ first — e.g. for $X \sim U(1,5)$, $\mathbb{P}(0 \le X \le 4) \neq \frac{4-0}{5-1}$; instead $\mathbb{P}(0 \le X \le 4) = \mathbb{P}(1 \le X \le 4) = \frac{4-1}{5-1} = \frac{3}{4}$.

Example: Uniform Distribution

Example

Let $X \sim U(1,5)$.
a) Find $\mathbb{P}(2 < X < 4)$.
Solution: $f_X(x) = \frac{1}{5-1} = \frac{1}{4}$ on $[1,5]$, so

$$\mathbb{P}(2 < X < 4) = \int_2^4 \frac{1}{4}\,\mathrm{d}x = \frac{1}{4} \cdot (4-2) = \frac{1}{2}.$$

Example

b) Find $c \in [1,5]$ such that $\mathbb{P}(3 < X < c) = \dfrac{1}{3}$.
Solution:

$$\frac{1}{3} = \int_3^c \frac{1}{4}\,\mathrm{d}x = \frac{1}{4}(c-3) \implies c - 3 = \frac{4}{3} \implies c = \frac{13}{3} \in [1,5].$$

Remark

Note that $\frac{1}{3} = \mathbb{P}(3 < X < c) = F_X(c) - F_X(3)$, so $F_X(c) = F_X(3) + \frac{1}{3}$; on a computer this could be solved for c using the quantile (inverse CDF) function.

The Exponential Distribution

Definition

Recall that the (discrete) Poisson random variable models the number of independent events occurring in a fixed interval of time. The **exponential distribution** is a continuous probability distribution that models the time between these independent events. It is commonly used to model waiting times.

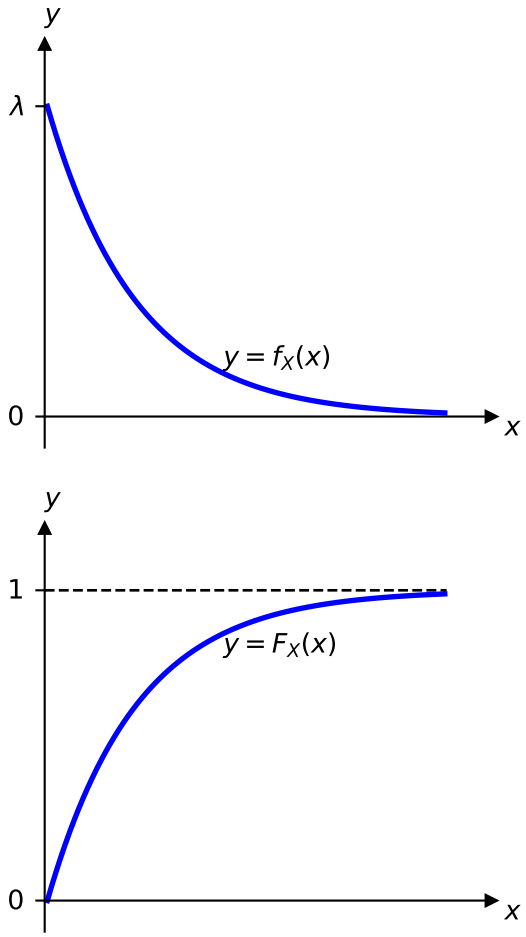


Figure 3: Graphs of PDF and CDF of $X \sim \text{Exp}(\lambda)$. Graphs are shown for $x \geq 0$ only. Both functions equal 0 for $x < 0$.

Definition

The PDF of the exponential distribution with parameter $\lambda > 0$ is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise,} \end{cases} \qquad F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Notation: $X \sim \text{Exp}(\lambda)$. The mean and variance are

$$\mathbb{E}(X) = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Example: Bus Arrivals (Exponential)

Example

Suppose the time between arrivals at a bus stop follows an exponential distribution with rate $\lambda = 0.05$ arrivals per minute.
a) Probability the next bus arrives within 10 minutes:

$$\mathbb{P}(X \leq 10) = F_X(10) = 1 - e^{-0.05 \cdot 10} = 1 - e^{-0.5} \approx 0.3935.$$

Example

b) Probability of waiting at least 15 minutes:

$$\mathbb{P}(X \geq 15) = 1 - F_X(15) = e^{-0.05 \cdot 15} = e^{-0.75} \approx 0.4724.$$

c) Average waiting time:

$$\mathbb{E}(X) = \frac{1}{0.05} = 20 \text{ minutes.}$$

The Normal Distribution

Remark

The **Normal distribution** is also known as the **Gaussian distribution**. Its PDF is symmetric, often called *bell-shaped*. It is widely used in probability and statistics, especially because of the **central limit theorem** (later in this course): averages of large samples behave "normally", regardless of the individual behaviour of the elements in the sample.

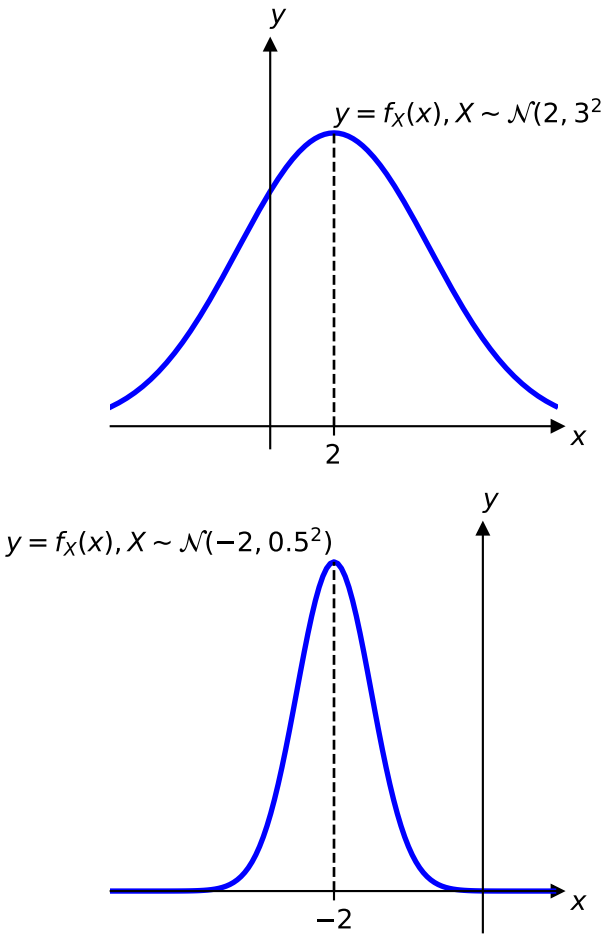


Figure 4: Graphs of PDF of $X \sim \mathcal{N}(2, 3^2)$ and $X \sim \mathcal{N}(-2, 0.5^2)$.

Definition

The normal distribution with mean $\mu \in \mathbb{R}$ and standard deviation $\sigma > 0$ is the continuous probability distribution with PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$, and $\mathbb{E}(X) = \mu$, $\mathrm{Var}(X) = \sigma^2$.

Remark

For example, $X \sim N(2, 9)$ means $\mu = 2$, $\sigma = 3$ (**not** $\sigma = 9$) — the second parameter is the variance σ^2.

Remark

Informally, $\mathbb{P}(X \in \Delta) \approx f_X(x) \cdot |\Delta|$ for a very short interval Δ of length $|\Delta| \to 0$, $x \in \Delta$.

The Standard Normal Distribution

Definition

The simplest case of a normal distribution is the **standard normal distribution** (or *unit normal distribution*), with $\mu = 0$, $\sigma = 1$. The PDF of $X \sim \mathcal{N}(0, 1)$ has special notation:

$$\varphi(x) := \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.$$

For $X \sim \mathcal{N}(\mu, \sigma^2)$ generally, $f_X(x) = \frac{1}{\sigma}\varphi(\frac{x-\mu}{\sigma})$.

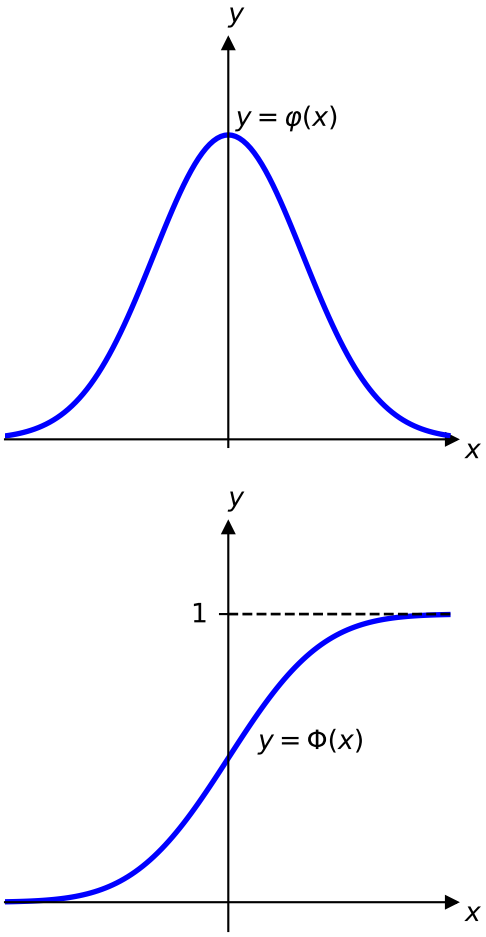


Figure 5: Functions $\varphi(x)$ and $\Phi(x)$

Definition

$$\Phi(x) := \int_{-\infty}^{x} \varphi(y)\,\mathrm{d}y = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{y^2}{2}}\,\mathrm{d}y.$$

This function cannot be expressed in terms of elementary functions — we use computer or statistical tables for its values.
It can be shown that if $X \sim \mathcal{N}(\mu, \sigma^2)$ then

$$F_X(x) = \Phi\Big(\frac{x-\mu}{\sigma}\Big) = \Phi(z), \qquad z = \frac{x-\mu}{\sigma}.$$

Also, $\Phi(-z) = 1 - \Phi(z)$.

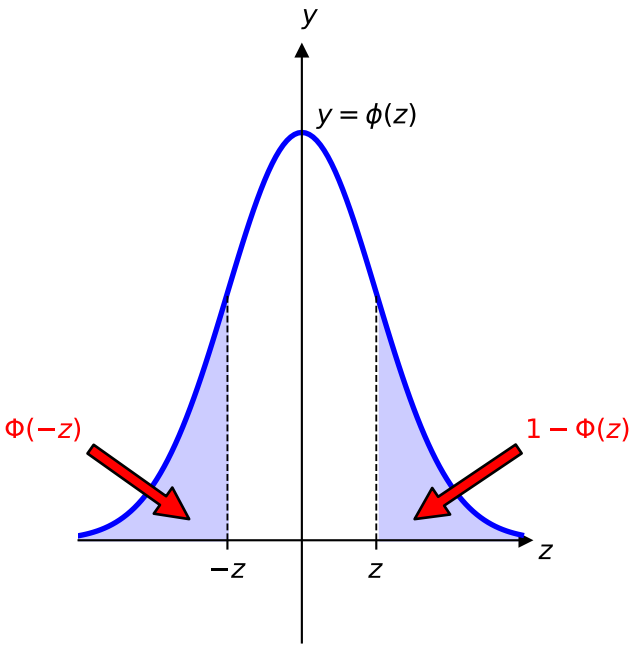


Figure 6: Illustration that $\Phi(-z) = 1 - \Phi(z)$

Using Standard Normal Tables

Statistical tables usually give $1 - \Phi(z)$ for $z \geq 0$. This immediately answers $\Phi(-z) = 1 - \Phi(z)$ and $\Phi(z) = 1 - (1 - \Phi(z))$.

$\frac{x-\mu}{\sigma}$	.00	.01	.02	.03	.04
0.0	.5000	.4960	.4920	.4880	.4840
0.1	.4602	.4562	.4522	.4483	.4443
0.2	.4207	.4168	.4129	.4090	.4052
0.3	.3821	.3783	.3745	.3707	.3669
0.4	.3446	.3409	.3372	.3336	.3300

Example

From the table:

$$1 - \Phi(0.63) = 0.2643 = \mathbb{P}(X > 0.63), \qquad 1 - \Phi(2.85) = 0.00219 = \mathbb{P}(X > 2.85).$$

Then $\Phi(-0.63) = 1 - \Phi(0.63) = 0.2643$, and $\Phi(0.63) = 1 - 0.2643 = 0.7357$.

Example: Heights (Normal Distribution)

Example

Calculate the probability that a randomly selected individual has a height between 162 cm and 169 cm, given that the population mean height is 165 cm and standard deviation 10 cm, with heights following a normal distribution.
Solution: $X \sim \mathcal{N}(\mu, \sigma^2)$, $\mu = 165$, $\sigma = 10$. We need $\mathbb{P}(162 \leq X \leq 169)$.

Example

Let $Z = \dfrac{X - \mu}{\sigma} \sim N(0, 1)$. Then

$$\mathbb{P}(162 \leq X \leq 169) = \mathbb{P}\left(\tfrac{162-165}{10} \leq Z \leq \tfrac{169-165}{10}\right) = \mathbb{P}(-0.3 \leq Z \leq 0.4) = \Phi(0.4) - \Phi(-0.3).$$

From the table, $1 - \Phi(0.4) = 0.3446$ and $1 - \Phi(0.3) = 0.3821$, so

$$\Phi(0.4) = 0.6554, \qquad \Phi(-0.3) = 0.3821,$$

$$\mathbb{P}(162 \leq X \leq 169) = 0.6554 - 0.3821 = 0.2733.$$

We can also check this with Python:

```
from scipy.stats import norm
norm.cdf(0.4) - norm.cdf(-0.3)
```

```
0.2733331637992768
```